MOHAMMED MARZUQ

Robotics and Automation

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I am a highly motivated and hands-on Robotics & IoT Engineer with a strong academic foundation in robotics and automation. My research internship at IIT Madras focused on drone technology for precision agriculture, where I gained valuable skills in autonomous systems, sensor integration (LIDAR, GPS, IMU), and control algorithms. Currently, I am working as an IoT Engineer at Akvo, where I design and deploy embedded systems using ESP32, GSM/Wi-Fi, and cloud dashboards. I am proficient in C and Python, and passionate about applying intelligent automation and real-time data systems in IoT-based applications. I am eager to contribute to innovative IoT and connected system solutions that enhance automation, safety, and efficiency.

Experience

Akvo Atmospheric Water Systems Pvt Ltd – Chennai *Feb 2025 – Present*
IoT Engineer

- Designed and implemented smart IoT systems for environmental and water monitoring using ESP32 microcontrollers.
- Integrated various sensors (INA219, DHT11/22, temperature) and communication modules (Wi-Fi, GSM) to enable real-time telemetry.
- Developed embedded firmware in C and Arduino IDE, ensuring energy-efficient operation with deep sleep modes and power monitoring.
- Implemented cloud dashboards using Blynk and Firebase for live data visualization, alerts, and remote control.
- Built robust OTA (Over-the-Air) firmware update systems, improving deployment and maintenance of remote devices.
- Contributed to remote diagnostic features and predictive maintenance by integrating real-time data with rule-based alert systems.

Advanced Manufacturing Technology Development Centre (IIT Madras Research Park)
Research Internship *(JAN) 2024 - (JUNE) 2024*

- Researched the application of VTOL drones in large-scale agriculture, focusing on precision irrigation, pest identification, and crop monitoring
- Analyzed advancements in sensor integration and autonomous navigation
- Identified cost and regulatory challenges, proposing solutions to improve adoption and sustainability in farming practices

PROJECTS

Smart IoT Energy Monitoring & Control System

- Designed and developed an IoT-based energy monitoring system using ESP32 and INA219 sensor to measure voltage, current, power, and energy in real-time.
- Integrated Blynk dashboard for live data visualization, remote monitoring, and system control.
- Implemented GSM (SIM800L) and Wi-Fi communication for dual-mode connectivity.
- Enabled OTA updates for remote firmware deployment and maintenance.
- Added features like alert notifications, data logging, and an energy reset button to enhance usability.
- Designed a custom PCB enclosure, optimized for compactness and airflow using dual cooling fans.
- Power-efficient design with sleep modes and real-time diagnostics to ensure system reliability.

DRONE FOR MONITORING FARMS
IIT Madras Research Park

- Implement machine learning algorithms for stable flight in varying conditions like wind, temperature, and terrain.
- Integrate multiple sensor inputs (LIDAR, GPS, IMU, cameras) for real-time data processing and decision-making.
- Use AI-based predictive algorithms for component monitoring, reducing downtime by alerting users to potential failures.
- Design control systems with redundancy to ensure safe landings and reliability in case of malfunctions.

DESIGN AND DEVELOPMENT THE GRIPPER FOR HANDLING SAND CORE IN DIECASTING INDUSTRIES

- Designed a specialized gripper mechanism to handle sand cores with curvy surfaces.
- Ensured safe and efficient pick-and-place of sand cores into mold cavities.
- Developed an adaptable gripper for various sand core shapes and sizes, enhancing precision.
- Reduced manual handling, addressing ergonomic issues and improving safety in the die casting industry.

PERSONAL ASSISTANT ROBOT

- Designed an Arduino-based Personal Assistant Robot for remote assistance.
- Integrated advanced robotic arm analysis for improved functionality.
- Aimed to assist elderly and mobility-impaired individuals in daily tasks.
- Focused on enhancing user independence and accessibility through automation.

DESIGN AND DEVELOPMENT OF TWIN CYLINDER LOW PRESSURE BOILER, (Diploma)

- Designed an Arduino-based Personal Assistant Robot for remote assistance.
- Integrated advanced robotic arm analysis for improved functionality.
- Aimed to assist elderly and mobility-impaired individuals in daily tasks.
- Focused on enhancing user independence and accessibility through automation.
- Studied a Lamont boiler model, essential for high-pressure applications in various industries.

Education

Diploma in Mechanical Engineering *2018 - 2021*
M.I.E.T Polytechnic college

B.E ROBOTICS AND AUTOMATION *2021 - 2024*
Sri Ramakrishna engineering college

Skills

- **Programming Languages:** Python, C
- **Robotics:** Robot Operating System (ROS)
- **Design Software:** Fusion 360
- **IoT Skills:** ESP32, Arduino, Wi-Fi/GSM, MQTT, Firebase, Blynk, OTAupdates, sensor integration.